

Zhixin Lu, Ph.D.

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Scientist II @ Allen Institute

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Google Scholar: <https://scholar.google.com/citations?user=ZnBE8XUAAAAJ&hl=en&oi=sra>

FOCUS

My research develops a **dynamical-systems and stochastic-process foundation for intelligence**, using it both to design new learning algorithms and architectures (*dynamical AI*) and to build AI that understands, predicts, and controls the dynamical world (*AI for dynamics*). I am especially drawn to **bio-inspired and physically-grounded computation**: architectures derived from how biological neural systems learn, and learning machines implementable in non-von Neumann physical substrates rather than only digital hardware. This endeavor connects foundational theory to modern architectures (e.g., transformers, diffusion models, SSMs, RNNs, Reservoir Computers, and world models) and to the neuroscience of how brains represent and forecast the dynamical world.

EMPLOYMENT & EDUCATION & INVITED VISITS

Allen Institute	Seattle, WA
Scientist II	2021 – present
Focus: designing AI that understands the dynamical world; designing dynamical AI; bio-inspired AI; dynamical system theory of learning and intelligence.	
Santa Fe Institute	Santa Fe, NM
Invited Speaker and Visiting Scientist	2026 May
Host: Yuanzhao Zhang	
Focus: Two underlying principles for sequential prediction from recurrent models and transformers.	
University of Pennsylvania, School of Engineering and Applied Science	Philadelphia, PA
Postdoctoral Researcher	2017 – 2021
Department of Bioengineering	
Supervisor: Dr. Danielle S. Bassett	
Focus: artificial intelligence; nonlinear dynamical systems; cognitive computational neuroscience.	
University of Maryland, Institute for Physical Science and Technology	College Park, MD
Ph.D. in Chemical Physics	2011 – 2017
Supervisor: Dr. Edward Ott	
Dissertation: “Dynamics of large systems of nonlinearly evolving units”	
Tsinghua University, Department of Physics	Beijing, China
Research Assistant	2010 – 2011
Supervisor: Prof. Yueheng Lan	
Focus: nonlinear dynamical systems.	
Shandong Normal University, School of Physics and Electronics	Shandong, China
B.S. in Physics	2006 – 2010

PUBLICATIONS

(>4000 citations)

Published / In Press

21. Calvin Yeung, Kris Ganjam, Zihan Zhang, Stefan Mihalas and **Zhixin Lu**, "A dataset of differentiable biologically-derived single neuron models." (Accepted by Cell Reports Methods).
20. **Zhixin Lu**, Kuśmierz, Łukasz, and Stefan Mihalas. "Identifying Stochastic Dynamics from Non-Sequential Data (DyNoSeD)." *Chaos: An Interdisciplinary Journal of Nonlinear Science* 36, no. 2 (2026).
19. Kim, Jason Z., Ling-Wei Kong, and **Zhixin Lu**. "SIGMa-DS: System identification from the geometric manifold of dynamical synchronization." *Chaos: An Interdisciplinary Journal of Nonlinear Science* 35, no. 10 (2025).
18. Kuśmierz, Łukasz, Ulises Pereira-Obilinovic, **Zhixin Lu**, Dana Mastrovito, and Stefan Mihalas. "Hierarchy of chaotic dynamics in random modular networks." *Physical Review Letters* 134, no. 14 (2025): 148402.
17. He, Xiaosong, Lorenzo Caciagli, Linden Parkes, Jennifer Stiso, Teresa M. Karrer, Jason Z. Kim, **Zhixin Lu**, Tommaso Menara, Fabio Pasqualetti, Michael R. Sperling, Joseph I. Tracy, and Dani S. Bassett. Uncovering the biological basis of control energy: structural and metabolic correlates of energy inefficiency in temporal lobe epilepsy. *Science Advances* 8, no. 45 (2022): eabn2293.
16. Kim, Jason Z., **Zhixin Lu**, Ann S. Blevins, and Dani S. Bassett. Nonlinear Dynamics and Chaos in Conformational Changes of Mechanical Metamaterials. *Physical Review X*. 2022 Mar 4;12(1):011042.
15. Smith, Lindsay M., Jason Z. Kim, **Zhixin Lu**, and Dani S. Bassett. Learning continuous chaotic attractors with a reservoir computer. *Chaos: An Interdisciplinary Journal of Nonlinear Science*. 2022 Jan 4;32(1):011101.
14. Qian, William, Lia Papadopoulos, **Zhixin Lu**, Keith A. Kroma-Wiley, Fabio Pasqualetti, and Dani S. Bassett. Path-dependent dynamics induced by rewiring networks of inertial oscillators. *Physical Review E*. 2022 Feb 9;105(2):024304.
13. Jason Z. Kim, **Zhixin Lu**, Erfan Nozari, George J. Pappas, and Danielle S. Bassett. Teaching recurrent neural networks to infer global temporal structure from local examples. *Nature Machine Intelligence*. 2020 Apr 19; 3(4):316-323.
12. **Zhixin Lu**, Danielle S. Bassett. Invertible Generalized Synchronization: A Putative Mechanism for Implicit Learning in Neural Systems. *Chaos: An Interdisciplinary Journal of Nonlinear Science*. 2020 Jun 16;30(6):063133.
11. Aaron F. Alexander-Bloch, Armin Raznahan, Simon N. Vandekar, Jakob Seidlitz, **Zhixin Lu**, Samuel R. Mathias, Emma Knowles, Josephine Mollon, Amanda Rodrigue, Joanne E. Curran, Harald H. Göring, Theodore D. Satterthwaite, Raquel E. Gur, Danielle S. Bassett, Gil D. Hoftman, Godfrey Pearlson, Russell T. Shinohara, Siyuan Liu, Peter T. Fox, Laura Almasy, John Blangero, David C. Glahn. Imaging Local Genetic Influences on Cortical Folding. *Proceedings of the National Academy of Sciences*. 2020 Mar 31;117(13):7430-6.
10. Zaixu Cui, Jennifer Stiso, Graham L. Baum, Jason Z. Kim, David R. Roalf, Richard F. Betzel, Shi Gu, **Zhixin Lu**, Cedric H. Xia, Xiaosong He, Rastko Ciric, Desmond J. Oathes, Tyler M. Moore, Russell T. Shinohara, Kosha Ruparel, Christos Davatzikos, Fabio Pasqualetti, Raquel E. Gur, Ruben C. Gur, Danielle S. Bassett, Theodore D. Satterthwaite. Optimization of energy state transition trajectory supports the development of executive function during youth. *Elife*. 2020 Mar 27;9:e53060.
9. **Zhixin Lu**, Jason Z. Kim, Danielle S. Bassett. Supervised chaotic source separation by a tank of water. *Chaos: An Interdisciplinary Journal of Nonlinear Science*. 2020 Feb 7;30(2):021101.
8. Jason Z. Kim, **Zhixin Lu**, Steven H. Strogatz, Danielle S. Bassett. Conformational control of mechanical networks. *Nature Physics*. 2019 Jul;15(7):714-20.

7. **Zhixin Lu**, Brian R. Hunt, Edward Ott. Attractor Reconstruction by Machine Learning. *Chaos: An Interdisciplinary Journal of Nonlinear Science*. 2018 Jun 22;28(6):061104.
6. Jaideep Pathak, Brian R. Hunt, Michelle Girvan, **Zhixin Lu**, Edward Ott. Model-free Prediction of Large Spatiotemporally Chaotic Systems from Data: A Reservoir Computing Approach. *Physical Review Letters*. 2018 Jan 12;120(2):024102.
5. Jaideep Pathak, **Zhixin Lu**, Brian R. Hunt, Michelle Girvan, Edward Ott. Using Machine Learning to Replicate Chaotic Attractors and Calculate Lyapunov Exponents from Data. *Chaos: An Interdisciplinary Journal of Nonlinear Science*. 2017 Dec 6;27(12):121102.
4. **Zhixin Lu**, Jaideep Pathak, Brian R. Hunt, Michelle Girvan, Roger Brockett, Edward Ott. Reservoir Observers: Model-free Inference of Unmeasured Variables in Chaotic Systems. *Chaos: An Interdisciplinary Journal of Nonlinear Science*. 2017 Apr 5;27(4):041102.
3. **Zhixin Lu**, Shane Squires, Edward Ott, Michelle Girvan. Inhibitory Neurons Promote Robust Critical Firing Dynamics in Networks of Integrate-and-fire Neurons. *Physical Review E*. 2016 Dec 19;94(6):062309.
2. **Zhixin Lu**, Kevin Klein-Cardeña, Steven Lee, Thomas M. Antonsen, Michelle Girvan, Edward Ott. Resynchronization of Circadian Oscillators and the East-West Asymmetry of Jet-Lag. *Chaos: An Interdisciplinary Journal of Nonlinear Science*. 2016 Sep 12;26(9):094811.
1. **Zhixin Lu**, Christopher Jarzynski, Edward Ott. Apparent Topologically Forbidden Interchange of Energy Surfaces under Slow Variation of a Hamiltonian. *Physical Review E*. 2015 May 15;91(5):052913.

Under Review / In Preparation

1. Harshil Sharma, Xiao-Ping Liu, Thomas Chartrand, Brian Kalmbach, Stefan Mihalas and **Zhixin Lu**. "The Unreasonable Effectiveness of Cell Types in Describing Neuronal Physiological Features." (In Preparation).

Non-peer reviewed

1. Thomas M. Antonsen, Michelle Girvan, **Zhixin Lu**, Edward Ott. Explaining the East/West Asymmetry of Jet Lag. *SIAM News*. 2017 Mar 1;50(2).

PRESENTATIONS

Invited Talks

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| "Synchronize or Hop: Two Mechanisms for Predicting the Dynamical World in Modern ML Models." | 2026 |
| Santa Fe Institute, Santa Fe, New Mexico | |
| "How recurrent neural networks infer dynamical models from data for prediction, inference, and source separation." | 2023 |
| APS March Meeting, Las Vegas, Nevada | |
| "Invertible Generalized Synchronization: A General Principle for Dynamical Learning in Neural Networks" | 2021 |
| Seminar at the Los Alamos National Laboratory's Advanced Network Science Initiative | |
| "Invertible Generalized Synchronization: A General Principle for Dynamical Learning in Neural Networks" | 2021 |
| Seminar at the Lanzhou Center for Theoretical Physics, Virtual | |
| "A putative mechanism for implicit learning in biological and artificial neural systems" | 2019 |
| Applied Dynamics Seminar, University of Maryland, College Park, MD | |

“Structural universality and Hebbian-like learning in an optimal context integration network” Summer Workshop on the Dynamics Brain, Friday Harbor, WA	2017
“Toward a theory of reservoir computing prediction” Applied Dynamics Seminar, University of Maryland, College Park, MD	2017
“Can topologically forbidden interchange of energy surfaces occur under slow variation of a Hamiltonian?” Informal Statistical Physics Seminar, IPST, University of Maryland, College Park, MD	2015

Conference Talks

“A First-Principles Methods for Learning Dynamics from Non-temporal Data” Cell Modeling in Space and Time, Cold Spring Harbor Laboratory (CSHL), NY	2026
“Inferring the Time-Averaged Jacobian Matrix of Wilson-Cowan Model with Machine Learning Method” SIAM Conference on Applications of Dynamical Systems (DS21), Online	2021
“Chaotic source separation solved by a tank of water through invertible generalized synchronization” APS March Meeting, Denver, CO	2020
“Why do reservoir computing networks predict chaotic systems so well” APS March Meeting, New Orleans, LA	2017
“Inhibitory neurons promote robust critical firing dynamics in networks of integrate-and-fire neurons” Dynamics Day, Silver Spring, MD	2017

Award-Winning Posters

“Resynchronization of circadian oscillators and the east-west asymmetry of jet-lag” 1 st prize/1 st place, IPST Spring Banquet, University of Maryland, College Park, MD	2016
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TEACHING EXPERIENCE

Students Advised, Allen Institute, Seattle, WA

Harshil Sharma, Undergrad @ University of Washington	2023 – Present
Calvin Yeung, PhD Student, Math and Computer Science @ UC Irvin Topics: The role of temporal complexity in neuronal computations	Summer 2023
Rachel Ding, Undergrad, Computation and Neural Systems @ Caltech Topics: Modeling single neuron dynamics using electrophysiological data	Summer 2022

Students Advised, University of Pennsylvania, Philadelphia, PA

Jason Z. Kim, PhD Student of Bioengineering Topics: reservoir computing; nonlinear dynamical systems and chaos	2017 – 2020
William Qian, Undergraduate Student of Physics and Computer Science Topic: path-dependent dynamics in networks of oscillators	2019 – 2020

Lindsay M. Smith, Undergraduate Student of Physics Topic: dynamical capacity of reservoir computers	2021 – 2022
Students Advised , University of Maryland, College Park, MD	
Steven Lee, Summer Undergraduate Student of Physics and Math Topic: modeling the dynamics of jet-lag recovery.	Summer ‘15
Kevin Klein-Cardena, Summer Undergraduate Student of Math Topic: modeling the dynamics of jet-lag recovery.	Summer ‘15
Teaching Assistant , University of Maryland, College Park, MD	
Introduction to Statistical Thermodynamics, Department of Physics	Fall ‘11
Principles of Modern Physics, Department of Physics	Fall ‘11
Classical Mechanics, Department of Physics	Spring ‘12
Differential Equations for Scientists and Engineers, Department of Mathematics	Fall ‘12
Applied Probability and Statistics I, Department of Mathematics	Spring ‘13
Chaotic Dynamics, Department of Physics	Fall ‘14
Teaching Assistant , Tsinghua University, Beijing, China	
Nonlinear Dynamics and Chaos, Department of Physics	Fall ‘10

HONORS AND AWARDS

Most downloaded and cited articles in CHAOS	2018
Most downloaded articles in CHAOS	2017
Best in Show Award, Summer Workshop on the Dynamic Brain, Friday Harbor, WA	2017
Travel Award, Dynamics Day, Silver Spring, MD	2017
1st Place Poster Prize, IPST Spring Banquet, University of Maryland, College Park, MD	2016
Merit Fellowship Award, University of Maryland, College Park, MD	2013
Dean’s Fellowship Award, University of Maryland, College Park, MD	2011
Second Prize in Chinese National Challenge Cup Technology Competition, Shandong, China	2009

MAJOR MEDIA COVERAGES

2022

12. “Quantifying abstraction in neural networks to increase understanding of human brain processing.”
By Anne Cockshott, **AIP Scilight**, January 4, 2022. Available at:
<https://aip.scitation.org/doi/10.1063/10.0009079>

2020

11. “Solving the Chaotic Source Separation Problem.”, By Meeri Kim, **AIP Scilight**, February 7, 2020. Available at: <https://aip.scitation.org/doi/10.1063/10.0000783>

2018

10. “Machine Learning’s ‘Amazing’ Ability to Predict Chaos.” By Natalie Wolchover, **Quanta Magazine**, April 18, 2018. Available at:
<https://www.quantamagazine.org/machine-learning-amazing-ability-to-predict-chaos-20180418/>
9. “A ‘perfect’ forecast? Machine learning may be the answer.” By Erin Wenckstern, **The Weather Network**, June 6, 2018. Available at:

<https://www.theweathernetwork.com/news/articles/artificial-intelligence-machine-learning-robots-weather-prediction-chaos-theory-mathematics-university-of-maryland/102924>

2016

8. “Unraveling the jet-lag asymmetry.” By Ashley G. Smart, **Physics Today**, October 1, 2016. Available at: <https://physicstoday.scitation.org/doi/10.1063/PT.3.3323>
7. “Why Jet Lag Can Feel Worse When You Travel from West to East.” By Joanna Klein, **The New York Times**, July 15, 2016. Available at: https://www.nytimes.com/2016/07/16/science/jet-lag-east-west.html?_r=0
6. “So That's Why Jet Lag Feels Worse When You Fly East.” By Suzy Strutner, **HuffPost**, July 15, 2016. Available at: https://www.huffpost.com/entry/jet-lag-circadian-rhythm_n_5788e090e4b0867123e0c6cd
5. “Study: Flying east is tougher on us than flying west.” By Arden Dier, **Fox News**, July 14, 2016. Available at: <https://www.foxnews.com/travel/study-flying-east-is-tougher-on-us-than-flying-west>
4. “Why your eastward jet lag is worse, according to math.” By Aria Hangyu Chen, **CNN**, July 13, 2016. Available at: <https://www.cnn.com/2016/07/13/health/eastward-jet-lag-worse/index.html>
3. “Researchers explain why jet lag can be worse when you travel east.” By Ben Guarino, **The Washington Post**, July 13, 2016. Available at: <https://www.washingtonpost.com/news/morning-mix/wp/2016/07/13/researchers-explain-why-jet-lag-can-be-worse-when-you-travel-east/>
2. “Scientists Break Down Why Flying East Is Worse for Jet Lag.” By Gillian Mohny and Dr. Imran Aslam, **ABC News**, July 13, 2016. Available at: <https://abcnews.go.com/Health/scientists-break-flying-east-worse-jet-lag/story?id=40521154>
1. “Mathematics could explain why jet lag seems worse when travelling east rather than west.” **ResearchGate News**, July 12, 2016. Available at: <https://www.researchgate.net/blog/post/mathematics-could-explain-why-jet-lag-seems-worse-when-travelling-east-rather-than-west>

SERVICE

Peer Reviewing

Reviewer for 15 journals: Physical Review Letters; Nature Machine Intelligence; Nature Communications; IEEE Transactions on Neural Networks and Learning Systems; Chaos; Journal of Nonlinear Science; Scientific Reports; SIAM Review; PLoS One; Neural Networks; Chaos, Solitons & Fractals; Earth System Dynamics; Entropy; Electronics; Philosophies.

Others

Invited panelist on panel discussion “The Ability of Artificial Intelligence Algorithms in New Computer Experiments to Tell the Future of Chaotic” in the 2019 Global Insurance Summit, New York City, NY. (<https://globalinsurancesummit2019.com/speakers/>)
